

Main paper: results tables

Table 1. Main real-data performance summary

This should be the core table.

Columns:

- Result block
- Metric
- Value

Rows:

- Regional InSAR posterior (R^2)
- Regional GRACE posterior (R^2)
- Regional SMAP posterior (R^2)
- Tiled InSAR posterior (R^2)
- Max ($\mathrm{ShallowLoad}$)
- Max ($\mathrm{DeepLoad}$)
- Max ($\mathrm{Groundwater}$)

Why:

- these are the main supported numbers of the grouped Bologna result.

Table 2. Independent wells validation summary

Columns:

- Validation block
- Count / statistic
- Value

Rows:

- evaluated stations
- positive best-match correlations
- stations with corr (≥ 0.3)
- stations with corr (≥ 0.5)

- median correlation
- top-10 mean correlation
- trusted stations
- trusted hydrogeologic groups

Why:

- this is the strongest external validation in the paper.

Table 3. Depth and best-state interpretation

Columns:

- Group
- Median correlation
- Notes

Rows:

- shallow wells
- intermediate wells
- deep wells

Then a second block or small companion table:

- Groundwater best state count
- ShallowLoad best state count
- DeepLoad best state count

Why:

- this helps interpretation and shows the grouped groundwater is strong but not the only physically relevant state everywhere.

Main paper: figures

Figure 1. Main grouped posterior result

Panels:

- ShallowLoad map

- DeepLoad map
- Groundwater map
- maybe one small panel with regional observed vs posterior timeseries

This should be the main visual result.

Figure 2. Wells validation summary panel

Panels:

- histogram or distribution of best correlations
- scatter / ranked correlations
- trusted subset summary
- maybe one panel for best-state counts

This is the strongest external-validation figure.

Figure 3. Trusted wells map

Show:

- Bologna domain
- wells
- trusted wells highlighted
- optionally hydrogeologic classes by symbol/color

This gives geographic credibility.

Figure 4. Representative station time series

Show about 4–6 representative wells:

- observed anomaly
- best matched grouped state
- chosen lag

This is very persuasive for reviewers.

Figure 5. Synthetic validation summary

One compact figure only:

- truth vs recovered summary

- deformation fit summary
- maybe one bar plot of synthetic (R^2) for (Sg), Load($\text{\text{total}}$), TWS, deformation

Keep it compact because synthetic is support, not the main result.

Supplementary Material: results tables

Table S1. Full data summary

Use the detailed dataset table from your workflow:

- InSAR
- W3RA
- GRACE
- SMAP
- SWOT
- wells
with coverage, variables, role, preprocessing.

Table S2. Conditioning diagnostics

Put the operator conditioning numbers here:

- old 5-layer patch
- old grouped patch
- current grouped multisensor Kalman
including ($\kappa(J)$), ($\kappa(\mathcal{I})$), numerical rank, scale spread.

This fits Appendix C.

Table S3. Full well-validation breakdown

Include:

- all trusted hydrogeologic groups
- station counts per group
- median / mean correlation per group
- lag statistics if useful

Table S4. SWOT overlap summary

Include:

- river rows
- lake rows
- river dates
- lake dates
- usable matched dates after fusion

This is useful because SWOT is in the paper, but not a major driver due to sparse overlap.

Supplementary Material: figures

Figure S1. Workflow diagram

A detailed workflow figure is ideal for the supplement if the main paper is tight.

Figure S2. Synthetic Stage 1 and Stage 2 detail panels

Include the extra synthetic comparisons you do not want in the main paper.

Figure S3. Conditioning diagnostics

Good candidates:

- singular value spectrum
- column norms
- correlation matrix of operator columns
- maybe observability Gramian spectrum

Figure S4. Depth-class summary

This belongs naturally in the supplement unless you need it in the main paper.

Figure S5. Lag histogram by state

Very good supplementary figure.

Figure S6. Trusted aquifer groups

Nice for interpretation, but supplementary is enough.

Figure S7. Top station timeseries panel

If the main paper only shows 4–6 stations, the larger panel can go here.

Figure S8. SWOT matched-date / exploratory result panel

Good to include because it documents that SWOT was tested, but sparse overlap limited its effect.

My strongest recommendation before Appendix D

Prepare these first:

Main paper

- Table 1 main performance
- Table 2 wells summary
- Figure 1 grouped posterior maps
- Figure 2 wells summary panel
- Figure 3 trusted wells map
- Figure 4 representative station time series

Supplementary

- Table S2 conditioning diagnostics
- Table S3 full well breakdown
- Figure S3 conditioning plots
- Figure S4 depth summary
- Figure S5 lag histogram
- Figure S8 SWOT exploratory panel

That is enough to let Appendix D be structured cleanly.